

Ahsanullah University of Science and Technology

Department of Computer Science and Engineering

Course Title: Cyber Security Lab | Course Number: CSE 4174

Assignment 1 — Version 1 (Odd Roll Numbers Only)

Scenario

You are building a **secure student portal** for AUST. Students must register with a password and log in using both their password and a one-time code from their phone.

Part A — Python Code

Note: No database is required. Store all user data in a Python dictionary in memory. Data does not need to persist after the program closes.

Write a Python program with the following features:

1. Student Registration

- Prompt for a username and password
- Enforce these password rules:
 - Minimum 8 characters
 - At least one uppercase letter
 - At least one number
- Hash the password using **bcrypt** and store it
- Generate a **TOTP secret key** for the user
- Display a **QR code** the user can scan with Google Authenticator
- Prevent duplicate usernames

2. Student Login

- Prompt for username, password, and TOTP code
- First verify the password using **bcrypt**
- Then verify the TOTP code using **pyotp**
- Both must pass for login to succeed
- After 3 failed attempts, block login for **20 seconds**

3. On Successful Login

- Display a welcome message
- Display the **last login time** of that user
- Update and store the new login time

Part B — Handwritten Explanation

Answer the following questions **by hand** on paper and submit with your code.

Q1. Registration — Run your program and register a user. Write down the exact bcrypt hash and the TOTP URI your program generated. Explain in 2 to 3 sentences what happened during registration.

Q2. Login — Log in with the correct password and correct TOTP code. Then try logging in with the correct password but a wrong TOTP code. What happened each time and why?

Q3. Account Lockout — Fail the login 3 times in a row. Write down what your program did and explain why this feature is important in a real system.

Q4. Last Login Time — Log in successfully twice. Write down what your program displayed as the last login time on the second login. Why is showing last login time useful for security?

Submission

Digital Submission:

- Upload your Python file named `[StudentID]_v1.py` to Google Classroom

Physical Submission:

- Submit a hard copy including:
 - Printed or handwritten Python code
 - Handwritten answers to Part B questions
 - A proper cover page with your Name, Student ID, Course Title, Course Number, and Assignment Number
- Hard copy must be submitted to your teacher in class

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